

Fruit Fly

Team: Cheesecake

Element E: Application of STEM Principles and Practices

Our boarding school residents need a way to remove fruit flies from their dormitories. This is because residents are irritated by the fruit flies and they could also pose a health risk. The fruit flies are attracted to a food item that was left in the dormitories a few weeks ago. Fruit flies have a short reproduction time so the quicker they are removed the easier it will be because there are less flies to remove. Quality of life in the dormitories would also improve if the fruit flies were to be removed.

The type of attractant we use will be dependent on if it is fragile to a range of temperatures as well as how much the repellent will need to be maintained. The odor of the repellent can change based on the chemical make up which is also correlated to how the repellent works on the target fruit flies. The repellent, of course, will also need to repel the target fruit flies. The odor of vinegar can be masked by a more pleasant smell while still attracting the target fruit flies.

The physical size of the prototype is closely determined by the material used because of the amount of bulk it could bring to the prototype. By using math for the measurements, we determined that the dimensions of the prototype were 7.5cm x 6.5cm x 10cm. This equals a volume of 487.5cm^3 . This is under our requirements of $1,000\text{cm}^3$.

The material we use can impact how fragile the prototype is and the chance of it breaking if it's touched or moved. Our design was constructed out of 3D printed material, specifically Polylactic Acid (PLA). This results in a sturdy prototype that will not break if touched, moved,

or if it falls. 3D material also allows the prototype to be under $1,000\text{cm}^3$ because of its characteristics of not being bulky and its strength that holds at any size of an object. PLA was chosen as opposed to another 3D printed material because it can hold any type of liquid and is what we had readily available.

Our two experts Mr. K. and Mr. S are qualified to review our solution because they both are in a position of authority regarding the students' dormitories. Also, Mr. K is an engineering teacher.